Food safety standards: FSSAI (Food Safety and Standards Authority of India), Functions, License, types of FSSAI License and compliance rules.
Food Hygiene: General principles of food microbiology and overview of foodborne pathogens, sources of microorganisms in the food chain (raw materials, water, air, equipment, etc.), Quality of foods, Microbial food spoilage and Foodborne diseases, Overview of beneficial microorganisms and their role in food processing and human nutrition, Food Analysis and Testing, General principles of food safety management systems, Hazard Analysis Critical Control Point (HACCP).

Food safety standards

What is FSSAI?

FSSAI stands for Food Safety and Standards Authority of India.
It is an autonomous body established under the Ministry of Health & Family Welfare, Government of India.
It was established under the Food Safety and Standards Act, 2006, which consolidates various acts and orders related to food safety.

Functions of FSSAI

- Framing regulations to lay down food safety standards.
- Laying down guidelines for accreditation of laboratories.
- Providing scientific advice and technical support to the government.
- Creating awareness and promoting food safety.
- Training programs for food business operators (FBOs).
- Collecting and analyzing data related to food consumption, contaminants, and emerging risks.
- Licensing and registration of food businesses.

FSSAI License

- Every food business operator (FBO) in India is required to obtain an FSSAI license or registration depending on the size and nature of the business.

Purpose of FSSAI License:

- Ensures that food products are tested and comply with food safety standards.
- Builds consumer trust.
- Legal requirement for running a food business.

Types of FSSAI Licenses

There are three main types of licenses based on the size and nature of the business:

a. FSSAI Basic Registration

For small FBOs with turnover less than ₹12 lakhs/year.

Suitable for small-scale food vendors, petty retailers, hawkers, etc.

b. FSSAI State License

For medium-sized FBOs with turnover between ₹12 lakhs to ₹20 crores/year.

Issued by the State Government.

Applicable to medium-sized manufacturers, storage units, transporters, marketers, etc.

c. FSSAI Central License

For large FBOs with turnover above ₹20 crores/year.

Also required if the business operates in more than one state or deals in imports/exports.

Issued by the Central Government.

Compliance and Rules

- FBOs must comply with FSSAI regulations to avoid penalties.

Key Compliance Requirements:

- Display of license at food premises.
- Regular hygiene and sanitation checks.
- Food safety audits and inspections.
- Proper labeling and packaging of products.
- Maintenance of records and documentation.
- Renewal of license before expiry.
- Timely reporting of any changes in the business.
- Penalties for Non-Compliance:
- Operating without a license can result in penalties up to ₹5 lakhs and/or imprisonment.
- FBOs can be suspended or shut down for serious violation

Structure of FSSAI

The structure of FSSAI includes a Chairperson appointed by the Central Government and 22 members, with one-third being women. It is supported by Scientific Committees and Panels for setting standards and a Central Advisory Committee for coordination with enforcement agencies. The responsibility for enforcement mainly lies with the State Food Safety Commissioners.

Food Hygiene

General principles of food microbiology

1. Microbial Growth

Microorganisms require specific conditions for growth and reproduction:
Nutrients,pH level, Temperature, Water activity (aw)
Understanding these factors helps in controlling microbial growth in food.

2. Types of Microorganisms

Common microbes in food include:
Bacteria, Yeasts, Molds, Viruses,
Each group behaves differently and impacts food safety and quality in unique ways.

3. Spoilage

Microorganisms cause spoilage by producing:
Enzymes, Acids, Toxins, This leads to undesirable changes in taste, texture, and appearance.

4. Food Safety

Pathogens such as: *Salmonella, Listeria, E. coli*
can cause foodborne illnesses through toxin production or direct infection.

5. Food Preservation Techniques

Used to inhibit or kill microbes:
Heat treatments (pasteurization, sterilization), Refrigeration & freezing,

Drying, Fermentation, Chemical preservatives

6. HACCP (Hazard Analysis and Critical Control Points)

A systematic preventive approach to:
Identify food safety hazards, Establish control measures, Ensure hazard prevention during production and processing

7. Quality Control

Ensures safety and hygiene through:
Microbiological testing (pathogens, spoilage organisms, hygiene indicators)

8. Regulatory Standards

Governed by national/international agencies:
Define acceptable microbial levels
Set safety and hygiene protocols

9. Emerging Issues

Modern challenges include:
Antimicrobial resistance
New food processing technologies
Global trade and novel food trends

10. Education and Research

Focus on:
Microbial behavior in food
Innovative preservation methods
Improving food safety systems

Overview of Foodborne Pathogens

Foodborne pathogens are microorganisms that can cause illness when consumed through contaminated food or water. They include:

Bacteria:*Salmonella*, *E. coli*, *Listeria*, *Campylobacter*

Cause infections via undercooked meat, raw eggs, unpasteurized milk, etc.

Viruses: *Norovirus*, *Hepatitis A*

Often spread through contaminated water or food handlers.

Parasites: Protozoa (e.g., *Giardia*, *Entamoeba*), roundworms, flatworms

Found in raw or undercooked meat, fish, and contaminated water.

Sources of Microorganisms in the Food Chain

Microorganisms can enter food from various points along the food chain:

Raw Materials: Carry soil-borne bacteria, fungi, or parasites.
Water: Used for washing or processing; may introduce *E. coli*, viruses.
Air: Airborne spores or dust can contaminate exposed food.
Equipment: Improperly cleaned surfaces spread pathogens.
Food Handlers: Poor hygiene leads to contamination from skin, respiratory tract, or feces.
Packaging Materials: Can harbor microorganisms if not sterile.

Quality of Foods & Microbial Food Spoilage

Spoilage: Caused by bacteria, molds, or yeasts through the production of gases, enzymes, acids, or toxins.

Indicators of spoilage: Off-smell, Sliminess, Gas formation, Discoloration, Spoiled food may not always be harmful, but it is undesirable and sometimes unsafe.

Foodborne Diseases

Diseases caused by ingestion of contaminated food or beverages:

Symptoms: Nausea, vomiting, diarrhea, fever, abdominal cramps.

Causes:
Pathogens (e.g., *Salmonella*, *Listeria*)
Toxins (e.g., *Staphylococcal enterotoxins*, mycotoxins)
Chemical hazards (pesticides, heavy metals)
Allergens (milk, nuts, gluten).

Overview of Beneficial Microorganisms

Not all microbes are harmful; some play positive roles in food and health:

In Food Processing:

Fermentation: *Lactobacillus* in yogurt, *Saccharomyces cerevisiae* in bread and alcohol. Improve shelf life, taste, texture

In Human Nutrition:

Probiotics: Support gut health, Boost immunity, Improve digestion.

Food Analysis and Testing

Essential for ensuring food quality and safety:

Types of Testing:

- **Microbiology Testing**: Detects pathogens and spoilage organisms.
- **Chemical Testing**: Identifies contaminants, additives, and nutrients.

- **Nutritional Analysis**: Verifies nutrient content for labeling.
- **Allergen Testing**: Identifies allergenic proteins.
- **Shelf-life and Sensory Evaluation**: Checks product stability and quality.

Techniques Used:

- Mass Spectrometry (MS)
- PCR (Polymerase Chain Reaction)
- ELISA (Immunoassays)
- NMR Spectroscopy
- Lateral Flow Devices (LFD).

General Principles of Food Safety Management Systems

Preventive Approach: Anticipates and controls hazards before they occur.

Good Manufacturing Practices (GMP): Ensures hygiene in food handling and processing.

Regular Audits and Inspections: Evaluate compliance with safety standards.

Education and Training: For staff to handle food safely.

Hazard Analysis Critical Control Point (HACCP)

A structured food safety system to control potential hazards.

Seven Principles:

Conduct hazard analysis
Identify critical control points (CCPs)
Establish critical limits
Monitor CCPs
Take corrective actions
Verify procedures
Document and record everything

Used globally, HACCP helps prevent biological, chemical, and physical food hazards at all stages of production